SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Software Engineering & Agile Development Methodologies

Department: CSE	**Course / Subject:** Software Engineering (CS601PC)
Semester: Semester 6	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: SDLC models, Agile Scrum framework, Requirement Engineering (SRS), Object-Oriented Analysis & Design (UML), Software Testing levels, and DevOps CI/CD pipelines.

Unit 1: Software Process Models

Waterfall model, Incremental process, Prototyping, Spiral model, Agile Manifesto principles, Scrum sprints, User Stories, and Kanban boards.

Unit 2: Software Requirements Engineering

Functional and non-functional requirements, Feasibility study, Requirement elicitation techniques, IEEE 830 SRS document standards, Use case specifications.

Unit 3: Architectural Design & UML

Modular design principles, Cohesion and Coupling, Architectural styles (Client-Server, Microservices), UML diagrams (Class, Sequence, Activity, Statechart).

Unit 4: Software Testing Strategies

Unit testing, Integration testing, System testing, Black-box testing (Equivalence Partitioning, Boundary Value Analysis), White-box testing (Basis Path, Cyclomatic Complexity).

Unit 5: Software Quality & DevOps

CMMI levels, ISO 9001 standards, Software metrics, Risk management, Git version control workflows, Continuous Integration and Continuous Deployment (CI/CD).